Fixed-Point Nash Equilibrium Search with Bayesian Stopping: A Functional-Analytic Approach to Explore-Exploit in Multi-Agent Systems

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Abstract

We develop a fixed-point theoretic approach to Nash equilibrium (NE) discovery in multi-agent systems. A Nash equilibrium is a fixed point of the joint best-response correspondence $\mathrm{BR} : \Delta \rightrightarrows \Delta$, where $\Delta = \prod_i \Delta(\mathcal{A}_i)$ is the product of strategy simplices. By smoothing BR with a temperature parameter $\tau > 0$, we obtain a continuous map $\mathrm{BR}_\tau : \Delta \to \Delta$ that is a contraction for sufficiently small τ (by Banach's fixed-point theorem). We then pose the problem of finding the *best* NE as a Bayesian optimization problem over the fixed-point set, with a Good-Turing-style estimator governing when to stop exploring. Key insight: *all selfish agents benefit from thorough NE search*, creating emergent cooperation in the exploration phase even in competitive games. This "Fixed-Point NE Search" (FP-NE) serves as a meta-algorithm wrapping the Ω-gradient, initializing it at the best discovered equilibrium.

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1 Setup

Consider an N-player normal-form game $\mathcal{G} = (N, \{\mathcal{A}_i\}_{i=1}^N, \{R_i\}_{i=1}^N)$ where agent i has action set $\mathcal{A}_i$ with $|\mathcal{A}_i| = d_i$ and payoff function $R_i : \mathcal{A} \to \mathbb{R}$, where $\mathcal{A} = \prod_i \mathcal{A}_i$.

Mixed strategies live on the product simplex $\Delta = \prod_{i=1}^N \Delta(\mathcal{A}_i)$, where $\Delta(\mathcal{A}_i) = \{\pi_i \in \mathbb{R}^{d_i}_{\geq 0} : \sum_a \pi_i(a) = 1\}$.

The expected payoff to agent i under joint mixed strategy $\pi = (\pi_1, \ldots, \pi_N)$ is:

$$V_i(\pi) = \sum_{a \in \mathcal{A}} \left(\prod_{j=1}^{N} \pi_j(a_j) \right) R_i(a). \tag{1}$$

Definition 1.1 (Best Response)**.** *The* best-response correspondence *for agent i is:*

$$\mathrm{BR}_i(\pi_{-i}) = \arg\max_{\pi_i \in \Delta(\mathcal{A}_i)} V_i(\pi_i, \pi_{-i}). \tag{2}$$

The joint best-response *is* $\mathrm{BR}(\pi) = (\mathrm{BR}_1(\pi_{-1}), \ldots, \mathrm{BR}_N(\pi_{-N}))$.

Definition 1.2 (Nash Equilibrium as Fixed Point)**.** *A strategy profile $\pi^* \in \Delta$ is a Nash equilibrium if and only if $\pi^* \in \mathrm{BR}(\pi^*)$, i.e., π^* is a* fixed point *of* BR.

2 Smoothed Best Response and Contraction

The best-response correspondence BR is set-valued and discontinuous, making fixed-point iteration problematic. We replace it with the *softmax (logit) best response*:

Definition 2.1 (Smoothed Best Response)**.** *For temperature $\tau > 0$, define the smoothed best-response map* $\mathrm{BR}_\tau : \Delta \to \Delta$ *by:*

$$\mathrm{BR}_{\tau,i}(\pi_{-i})(a) = \frac{\exp\left(\frac{1}{\tau} \sum_{a_{-i}} \pi_{-i}(a_{-i}) R_i(a, a_{-i})\right)}{\sum_{a'} \exp\left(\frac{1}{\tau} \sum_{a_{-i}} \pi_{-i}(a_{-i}) R_i(a', a_{-i})\right)}. \tag{3}$$

The key property: BR_τ is a smooth self-map on Δ, and its fixed points are the *quantal response equilibria* (QRE) of McKelvey and Palfrey (1995). As $\tau \to 0$, QRE converge to Nash equilibria.

Theorem 2.2 (Contraction Property)**.** *For any finite game with payoffs bounded by $\|R_i\|_\infty \leq M$, the smoothed best-response map* BR_τ *satisfies:*

$$\| \mathrm{BR}_\tau(\pi) - \mathrm{BR}_\tau(\pi') \|_\infty \leq \frac{M}{\tau} \cdot \|\pi - \pi'\|_\infty \tag{4}$$

for all $\pi, \pi' \in \Delta$. In particular, if $\tau > M$, then BR_τ *is a* contraction mapping *with Lipschitz constant $c = M/\tau < 1$.*

Proof. The softmax function $\sigma_\tau(x)(a) = \exp(x_a/\tau)/\sum_{a'} \exp(x_{a'}/\tau)$ has Lipschitz constant $1/(2\tau)$ in $\|\cdot\|_\infty$ norm. The linear map $\pi_{-i} \mapsto R_i(\cdot, \pi_{-i})$ has operator norm bounded by M. By composition, $\mathrm{BR}_{\tau,i}$ is Lipschitz with constant M/τ. □

Corollary 2.3 (Banach Fixed-Point Theorem Applied)**.** *When $\tau > M$, the damped iteration $\pi_{t+1} = (1-\alpha)\pi_t + \alpha \cdot \mathrm{BR}_\tau(\pi_t)$ converges to the unique QRE at geometric rate* $\max(1-\alpha, c)^t$ *from any initialization.*

Remark 2.4. *For $\tau \leq M$,* BR_τ *may have multiple fixed points (corresponding to multiple QRE/NE). This is precisely the regime where the* equilibrium selection problem *arises and where our Bayesian search becomes valuable.*

3 Bayesian NE Discovery

When τ is small (close to NE), the game may have multiple fixed points. We frame the problem of finding the *best* fixed point as a species-discovery problem.

3.1 The Explore-Exploit Tradeoff

Definition 3.1 (NE Search Problem)**.** *Given a game $\mathcal{G}$ with unknown set of Nash equilibria $\mathcal{E} = \{\pi^{*,1}, \ldots, \pi^{*,K}\}$ (where K is unknown), find $\pi^{*,k}$ maximizing a selection criterion $\phi(\pi^{*,k})$ (e.g., social welfare $\sum_i V_i(\pi^{*,k})$, or agent i's payoff $V_i(\pi^{*,k})$).*

Each "search" consists of:

1. Sample a random initialization $\pi_0 \sim \mathrm{Dir}(\mathbf{1})$ (or perturb a known NE).
2. Run damped BR_τ iteration until convergence.
3. Record the fixed point (if residual $\| \mathrm{BR}_\tau(\pi) - \pi \| < \epsilon$).
4. Check if it is distinct from previously discovered NE.

3.2 Good-Turing Estimator for Total NE Count

We adapt the classical species-discovery framework to NE counting. After n searches discovering d distinct NE, the probability of finding a *new* NE on the next search is estimated by:

$$\hat{p}_{\text{new}} = 1 - \hat{C}_n, \qquad \hat{C}_n = 1 - \frac{f_1}{n} \tag{5}$$

where f_1 is the number of NE discovered exactly once (singletons) and $\hat{C}_n$ is the Good-Turing coverage estimator.

The Chao1 estimator for total species (NE) count:

$$\hat{K} = d + \frac{f_1^2}{2f_2} \tag{6}$$

where f_2 is the number of NE discovered exactly twice.

3.3 Bayesian Stopping Rule

Definition 3.2 (Stopping Criterion)**.** *Define the* exploration value *for agent i as:*

$$EV_i = \mathbb{P}(\exists \text{ undiscovered } \pi^* : V_i(\pi^*) > V_i^{best}) \tag{7}$$

where $V_i^{best} = \max_{k \le d} V_i(\pi^{*,k})$ is the best discovered payoff for agent i.

Stop exploring when $EV_i < \delta$ for all agents i.

The probability decomposes as:

$$\mathrm{EV}_i \le \mathbb{P}(\text{undiscovered NE exist}) \times \mathbb{P}(\text{random NE is better} \mid \text{it exists}). \tag{8}$$

The first factor is estimated by $1 - d/\hat{K}$. The second factor can be bounded using the empirical distribution of discovered NE payoffs.

4 Cooperation from Selfish Search

The central insight: *in games with multiple NE, the search for the best NE is a public good that selfish agents voluntarily provide.*

Proposition 4.1 (Search as Public Good)**.** *Consider a game with NE set* $\mathcal{E} = \{\pi^{*,1}, \ldots, \pi^{*,K}\}$ *where* $K \geq 2$*. Let* $\pi^{*,P}$ *denote the Pareto-best NE (maximizing* $\sum_i V_i$*). If agents use a selection criterion* $\phi_i(\pi) = V_i(\pi)$*, then under mild conditions on the payoff structure*[1]*, increasing the search budget benefits* all *agents:*

$$\frac{\partial}{\partial n}\mathbb{E}\left[\max_{k \leq d(n)} V_i(\pi^{*,k})\right] > 0 \quad \forall i. \tag{9}$$

Sketch. More search discovers more NE. Across NE, payoffs tend to be positively correlated (Pareto-better NE are better for most agents). Thus the probability that a newly discovered NE improves agent i's best available payoff is positive for all i simultaneously. □

Remark 4.2. *This is a* functional-analytic *grounding of cooperation: it arises not from altruism or mechanism design but from the topological structure of the fixed-point set. The fixed points of* BR *are shared landmarks that all agents want to map.*

5 Integration with the Ω-Framework

FP-NE search serves as a *Phase 0* preceding the Ω-gradient:

Algorithm 1: Ω-PG with FP-NE Initialization

Input: Game $\mathcal{G}$, Ω-config $(\gamma, w, \lambda, \beta)$, FP-NE budget B, confidence δ
Output: Learned policy profile $\hat{\pi}$

```
// Phase 0:  Fixed-Point NE Search
```

Initialize $\mathcal{D} \leftarrow \emptyset$; `// discovered NE`
for $s = 1, \ldots, B$ **do**
  Sample $\pi_0 \sim \mathrm{Dir}(\mathbf{1})$ or perturb best known NE;
  $\pi^* \leftarrow \mathrm{DampedBR}_\tau(\pi_0)$; `// Banach iteration`
  if $\|\mathrm{BR}_\tau(\pi^*) - \pi^*\| < \epsilon$ ***and*** π^* *is new* **then**
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{\pi^*\}$;
  end
  if $EV_i < \delta$ *for all* i **then**
    break ; `// Bayesian stopping`
  end
end
$\hat{\pi}_{\text{init}} \leftarrow \arg\max_{\pi^* \in \mathcal{D}} \sum_i V_i(\pi^*)$;

```
// Phase 1:  Ω-Gradient from best NE
```

$\pi_0 \leftarrow \hat{\pi}_{\text{init}}$;
for $n = 1, 2, \ldots$ **do**
  $\pi_{n+1} \leftarrow \mathrm{proj}_\Delta(\pi_n + \gamma_n \cdot w_n \cdot (\hat{v}_n + \lambda_n \cdot \mathrm{OS}_n + \beta_n \cdot \mathrm{coop}_n))$;
end
return π_n

Theorem 5.1 (FP-NE Preserves Ω-Convergence)**.** *If the Ω-gradient converges to a stable NE* π^* *from any initialization in basin* $\mathcal{B}(\pi^*)$*, and FP-NE search discovers* π^* *with probability* $\geq 1 - \epsilon$*, then Ω-PG with FP-NE initialization converges to the best discoverable NE with probability* $\geq 1 - \epsilon$*.*

The advantage over random initialization: random init lands in the basin of an arbitrary NE. FP-NE search explicitly maps the fixed-point landscape and selects the best basin before starting the Ω-gradient.

[1] Specifically, when the correlation between agents' payoffs across NE is positive: $\mathrm{Corr}(V_i(\pi^{*,k}), V_j(\pi^{*,k})) > 0$ over k.

6 Connection to Kakutani and Brouwer

The existence of Nash equilibria is guaranteed by Kakutani's fixed-point theorem (for the set-valued BR correspondence) or equivalently by Brouwer's theorem (for the smoothed map BR_τ).

Theorem 6.1 (Nash 1950, via Kakutani)**.** *Every finite game has at least one mixed-strategy Nash equilibrium.*

Our contribution is algorithmic: given that fixed points *exist*, we provide an efficient method to *find them all* (or at least the best ones) with Bayesian confidence guarantees on completeness.

The functional-analytic perspective:

- Δ is compact and convex (finite-dimensional simplex).
- $\mathrm{BR}_\tau : \Delta \to \Delta$ is continuous for $\tau > 0$.
- By Brouwer, BR_τ has at least one fixed point.
- For $\tau > M$: unique fixed point (Banach contraction).
- For $\tau \leq M$: possibly multiple fixed points. Decreasing τ from $> M$ to near 0 traces the *QRE correspondence* connecting the unique high-τ QRE to the (possibly multiple) NE.

This suggests a *homotopy method*: start at high τ (unique contraction), gradually decrease τ, and track how the unique fixed point bifurcates into multiple NE. Branch points in this homotopy correspond to equilibrium selection events.

7 Experimental Predictions

1. **NE Discovery**: FP-NE search discovers all NE in games with known equilibrium sets (validated against nashpy).
2. **Bayesian Stopping**: The stopping criterion correlates with actual search completeness.
3. **Equilibrium Selection**: FP-NE + Ω-PG achieves higher social welfare than Ω-PG alone in games with multiple NE.
4. **Contraction Rates**: Empirical contraction constants of BR_τ match the theoretical bound M/τ.
5. **Cooperation from Selfishness**: In multi-NE games, increasing search budget improves welfare for *all* agents.

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